**DRILL 09**

WATER SUPPLY

NFPA 1001 Level 1 (JPR 4.3.14, 4.3.15) / WorkSafeNB Reg 91-191

# LARGE DIAMETER HOSE (LDH) & HYDRANT / SUPPLY LAYS

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Hydrant Firefighter & Driver/Apparatus Crew

## 🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to execute a **Forward Supply Lay** using 4-inch or 5-inch (100mm/125mm) Large Diameter Hose (LDH) with sexless Storz couplings, connect to a pressurized municipal hydrant or dry hydrant, flush the hydrant, make the connection with a hydrant gate valve, and charge the supply line without water hammer.

## 📦 TOOLS & APPARATUS SETUP

- Pumper with LDH supply bed (minimum 200–300 ft loaded).
- Hydrant kit/bag: Hydrant wrench, 2.5" gate valve, 4"/5" Storz-to-steamer adapter, rubber mallet, spanner wrenches.
- Station training hydrant or simulated hydrant marker.

## ⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Snag & Flail Hazard:** When laying LDH from a moving apparatus, ensure the hydrant firefighter stands firmly in the safety zone (behind the hydrant or off the road curb) and never steps into the loop of paying-out hose.

⚠️ **Water Hammer:** Never open or close a hydrant or master intake gate rapidly. Minimum 30 seconds to open; minimum 30 seconds to close!

⚠️ **Storz Locking Latch:** Ensure the locking lever on the Storz coupling snaps shut. If not locked, the coupling can twist and decouple violently under 150 PSI!

🛑 **THE STOP RULE: Shout "STOP!" immediately if the paying-out hose catches on a firefighter's boot or gear.**

## PHASE 1: THE HYDRANT BAG & STORZ COUPLING MECHANICS (10 MINUTES) — SLOW DRY RUN

1. **The Storz Advantage:** Sexless couplings (quarter-turn twist). Align the locking lugs, push firmly together, and rotate 1/3 turn until the spring-loaded lock clicks.
2. **The Hydrant Kit Inventory:**
   - Hydrant wrench (pentagonal socket).
   - 4" or 5" Storz-to-thread adapter.
   - 2.5" gate valve (allows a second line to be connected later without shutting the hydrant down).

# DRILL 09: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Water Supply • Practical Application

## PHASE 2: THE FORWARD LAY (CATCHING THE HYDRANT) (20 MINUTES) — BUDDY COACHING

1. **Dismounting & Wrapping the Hydrant:**
   - Apparatus stops 15–20 feet past the hydrant.
   - Hydrant firefighter steps off apparatus with hydrant bag, grasps LDH coupling, and pulls 10–15 feet of hose off the tailboard.
   - Wrap the hose completely around the barrel of the hydrant in a secure loop.
   - Stand with a boot firmly planted on the crossed hose to anchor it against the ground.
   - Signal the driver with a loud voice and raised arm: *"LAY THE LINE! GO GO GO!"*
   - As the apparatus drives forward at 8–10 km/h, the hose pays out of the bed smoothly.
2. **Hydrant Preparation & Connection:**
   - Unwrap the hose from the hydrant barrel.
   - Remove the steamer port cap using the hydrant wrench.
   - **Flush the Hydrant:** Crack the operating nut on top 2–3 turns until water flows clear of gravel and rust, then close.
   - Thread the 4"/5" Storz adapter onto the steamer port and tighten with spanner.
   - Thread a 2.5" gate valve onto one of the side ports (closed position).
   - Connect the LDH Storz coupling to the adapter; verify locking latch snaps shut.

## PHASE 3: CHARGING THE SUPPLY LINE & AIR BLEEDING (15 MINUTES) — TIMED / STRESS

1. Await radio confirmation from the pump operator: *"Engine 1 is ready for hydrant water. Charge the supply line!"*
2. Rotate the hydrant top operating nut counter-clockwise slowly until fully open (feel for the valve to hit its stop, then back off 1/4 turn to prevent sticking).
3. Pump operator opens the pump intake bleeder valve to exhaust air before opening the master intake valve.

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### ⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Stepping inside the hose loop:* If the hose snags on the tailgate while the truck drives away, an internal loop will whip around your legs and cause severe fractures. Stay outside the loop!
- ❌ *Not flushing the hydrant:* Rocks, sediment, and rust sitting inside the hydrant barrel will shoot into the fire pump, destroying the impeller and packing seals.
- ❌ *Closing the hydrant fast:* Slamming a hydrant shut creates a catastrophic pressure wave (water hammer) that can rupture municipal mains under the street!

### 🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the Storz lock snap closed properly?", "How long did it take between dismounting the truck and having the hydrant connected?"***

### ✅ APPARATUS RESET CHECKLIST

- ☐ Hydrant shut completely until operating nut stops; check barrel drainage (suction check).
- ☐ Caps replaced and tightened hand-tight with wrench.
- ☐ LDH drained: Walk the hose with a drainage roller or over-the-shoulder walk.
- ☐ LDH repacked in the apparatus bed (flat load with folds slightly staggered).
- ☐ Hydrant kit repacked and returned to apparatus compartment.
- ☐ Attendance roster signed.

| DRILL LEADER SIGNATURE | SAFETY OFFICER SIGNATURE | DATE COMPLETED | TRAINING HOURS |
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